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Cross-Frequency Adaptive Synergistic Mamba for Hyperspectral Image Classification

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***Abstract*—Although Mamba networks based on State Space Models (SSMs) can efficiently model long-range dependencies with linear complexity, their mandatory 1D serialized input severely disrupts the original 2D spatial topology when processing hyperspectral images (HSI), leading to a loss of directional perception for geometric edges and smooth backgrounds. To address this bottleneck, a Cross-Frequency Adaptive Synergistic Mamba (CFAS-Mamba) network is proposed, which effectively decouples spatial features into frequency-domain to construct feature subspaces with absolute physical orientation attributes. Specifically, a Wavelet-Guided Anisotropic Serialization (WGAS) module is developed to impose directional receptive fields based on the physical properties of different frequency bands, enabling the precise and lossless capture of edge details in horizontal, vertical, and diagonal directions. Furthermore, considering the high sensitivity of high-frequency features to sensor noise, a Cross-Frequency Synergistic Attention (CFSA) mechanism is introduced to achieve noise-robust collaborative evolution between high- and low-frequency features. Combined with a progressive hierarchical pyramid architecture, CFAS-Mamba extracts highly discriminative multiscale spatial-spectral representations. Comprehensive evaluations on four benchmark datasets demonstrate that the proposed method significantly outperforms existing state-of-the-art (SOTA) networks in terms of both classification accuracy and generalization capability.**

***Index Terms*—Hyperspectral image classification, Mamba network, Frequency-domain decoupling, Attention fusion**

I. Introduction

HYPERSPECTRAL images (HSIs) [1] consist of dozens to hundreds of contiguous spectral bands, encoding both intricate spatial structures and abundant spectral information. Spatial information precisely delineates geometric attributes such as structure, shape, and texture, whereas spectral information characterizes the physical and chemical properties of surface materials [2]. This dual nature endows HSIs with the unique capability to discriminate and analyze complex targets, driving their widespread adoption in remote sensing prospecting [3], intelligent agriculture [4], environmental monitoring [5], and related domains.

HSI classification aims to assign each pixel to a specific land-cover category. Early research predominantly employed conventional machine learning methods for feature extraction, such as principal component analysis (PCA) [6], support vector machines (SVM) [7], [8], sparse representation (SR) [9]–[11], and linear discriminant analysis (LDA) [12], [13]. Despite their theoretical soundness, these traditional classifiers suffer from two critical limitations in practice: (1) their performance depends heavily on hand-crafted priors and extensive feature engineering, and (2) they typically exploit only a subset of spectral bands while neglecting the rich spatial contextual information inherent in HSIs.

In contrast to traditional classifiers, deep learning (DL)-based methods [14]–[18], including convolutional neural networks (CNNs) [19]–[21], recurrent neural networks (RNNs) [22]–[24], and stacked autoencoders (SAEs) [25], have achieved remarkable success in visual tasks owing to their powerful fitting capacity and automated feature extraction. Through hierarchical non-linear transformations, DL approaches learn high-level abstract representations directly from raw data, eliminating the need for labor-intensive manual feature design. Among these DL approaches, CNNs [26]–[28] have emerged as a dominant force in HSI classification research, leveraging localized receptive fields and weight sharing to extract both spectral contexts and spatial features.

Nevertheless, the intrinsic locality of convolutional kernels confines CNN-based methods to a restricted square receptive field, limiting their capacity to model long-range dependencies across the entire HSI scene. To overcome this structural bottleneck, the Vision Transformer (ViT) [29] introduced a paradigm shift. By virtue of the multi-head self-attention (MHSA) mechanismf, Transformers [30]–[33] attend to all spatial positions of the input sequence, thereby establishing comprehensive global interactions unconstrained by local receptive fields. Building upon this foundation, recent works have explored hybrid architectures to further enhance spatial-spectral feature extraction. Fang *et al.* [34] integrated deformable convolutions with a hierarchical Transformer to capture both fine-grained details and global structures. Sun *et al.* [35] devised a dual-branch framework enabling deep interaction between CNN and Transformer streams. Zhao *et al.* [36] incorporated hypergraph neural networks into Transformers to model high-order relational structures alongside long-range dependencies.

Despite their representational power, Transformer architectures incur a fundamental computational burden: the quadratic complexity $\mathcal{O}(N^2)$ of the self-attention mechanism with respect to sequence length imposes prohibitive memory and computational costs, which are particularly severe when processing the HSIs. In sharp contrast, the Mamba architecture [37]–[39], which is grounded in structured SSMs, establishes long-range dependencies through state transitions with linear

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computational complexity $\mathcal{O}(N)$. Its input-dependent selective scan mechanism dynamically gates state updates, adaptively suppressing irrelevant computations. By encoding input data as a sequence of latent states, Mamba naturally captures long-range dependencies inherent in sequential data, demonstrating substantial promise for efficient high-dimensional HSI processing.

Recently, several pioneering efforts have explored the application of Mamba to HSI classification. Shi *et al.* [40] proposed an enhanced 3D spectral-spatial Mamba integrated with dendritic learning to mitigate the loss of local details induced by Mamba's sequential compression. Recognizing that standard Mamba models predominantly operate at a single scale, Liao *et al.* [41] introduced a hierarchical state space framework with parallel multi-scale convolutional embeddings. To overcome the rigidity of a single state evolution pattern, Zhao *et al.* [42] devised a multi-pattern state evolution mechanism that dynamically adapts to heterogeneous feature distributions.

Despite these notable advances, existing Mamba-based HSI classification networks still face two fundamental yet unresolved bottlenecks. First, Mamba's mandatory 1D sequential input format inevitably disrupts the intrinsic 2D spatial topology of HSI patches, causing an irreversible loss of directional edge information. This is particularly detrimental for delineating fine-grained land-cover boundaries and preserving geometric integrity in heterogeneous scenes. Second, the standard serialization paradigm treats all spatial locations uniformly, failing to discriminate between structurally informative high-frequency components (e.g., edges and textures) and noise-corrupted high-frequency artifacts, which are especially pronounced in HSI due to sensor-induced noise. Collectively, these limitations severely constrain the representational fidelity of Mamba-based models.

To address the above problems, a CFAS-Mamba network is proposed, which innovatively decouples spatial-spectral features into frequency-domain to construct feature subspaces with explicit physical orientation attributes. By integrating wavelet-guided anisotropic serialization with cross-frequency synergistic attention within a progressive hierarchical pyramid architecture, CFAS-Mamba facilitates joint spatial-spectral learning with intrinsic noise mitigation, orientation-adaptive serialization, and pyramidal multi-scale abstraction for HSI classification. The main contributions are summarized as follows:

1) A CFAS-Mamba network is proposed for HSI classification, which decouples spatial-spectral information into frequency-domain. By organizing WGAS and CFSA modules within a progressive hierarchical pyramid architecture, it yields anisotropic, noise-suppressed and multi-scale spatial–spectral representations.
2) A WGAS module is developed to construct frequency-specific feature subspaces with explicit physical orientation attributes. Through channel-wise Stationary Wavelet Transform (SWT) and frequency directional scanning mechanism, WGAS yields lossless conversion from 2D spatial layouts to 1D token sequences while preserving critical anisotropic geometric boundaries.
3) A CFSA module is introduced to achieve collaborative evolution between disparate high- and low-frequency components. Leveraging the noise-robust low-frequency topology as a semantic anchor, CFSA employs a context-gated mechanism to adaptively suppress high-frequency noise, and subsequently implements a cross-orientation attention fusion mechanism across orthogonal gradient subbands to synthesize high-order spatial-spectral textures.

The remainder of this paper is organized as follows. Section II delineates the specifics of the proposed CFAS-Mamba architecture. Section III presents comprehensive experimental evaluations and comparative analyses against state-of-the-art methods. Finally, Section IV concludes this study.

II. Proposed method

A. Overall Framework

Given a HSI patch denoted as $\mathbf{X} \in \mathbb{R}^{B\times C\times H\times W}$, where B is the batch size, C represents the number of contiguous spectral bands, and $H \times W$ is the spatial extent, the proposed CFAS-Mamba aims to construct a non-linear parameterized mapping $\mathcal{F}_\theta : \mathbb{R}^{C\times H\times W} \to \mathbb{R}^K$, where K denotes the total number of land-cover categories. The end-to-end pipeline is formalized as the composition of four stages:

$$\hat{\mathbf{y}} = \Phi_{\text{Mamba}} \circ \Phi_{\text{CFSA}} \circ \Phi_{\text{WGAS}} \circ \Phi_{\text{SSJP}}(\mathbf{X}) \tag{1}$$

where Φ_{SSJP}, Φ_{WGAS}, Φ_{CFSA}, and Φ_{Mamba} denote the Spectral-Spatial Joint Processing (SSJP), WGAS, CFSA, and Mamba Encoder, respectively. The network architecture, as illustrated in Fig. 1, enforces deep spectral-spatial coupling through a frequency-domain unification framework. Specifically, the SSJP module performs inter-band spectral recombination via pointwise convolutions to project raw vectors into a learned embedding space, followed by depthwise convolutions for per-channel spatial context aggregation. The WGAS module then applies channel-wise SWT to decompose each spectral embedding map into four physically oriented frequency subbands and employs frequency-adaptive directional scanning to flatten each subband into 1D token sequences with customized anisotropic trajectories that respect both wavelet-derived orientation priors and inter-channel spectral correlations. The CFSA module subsequently exploits the low-frequency subband to adaptively gate and denoise high-frequency components, and establishes cross-orientation attention fusion to reconstruct intact high-order spectral-spatial textures. Finally, the Mamba encoder processes the fused multi-frequency sequence via input-dependent selective SSM with $\mathcal{O}(N)$ linear complexity, capturing global long-range dependencies while preserving spatial-spectral discriminability. A progressive hierarchical pyramid is constructed by stacking WGAS-CFSA-Mamba stages, enabling systematic extraction of multi-scale representations from low-level frequency-domain textures to high-level semantic abstractions.

B. Spectral-Spatial Joint Embedding Module

HSIs contain a high degree of spectral redundancy. The densely sampled contiguous bands exhibit strong inter-band

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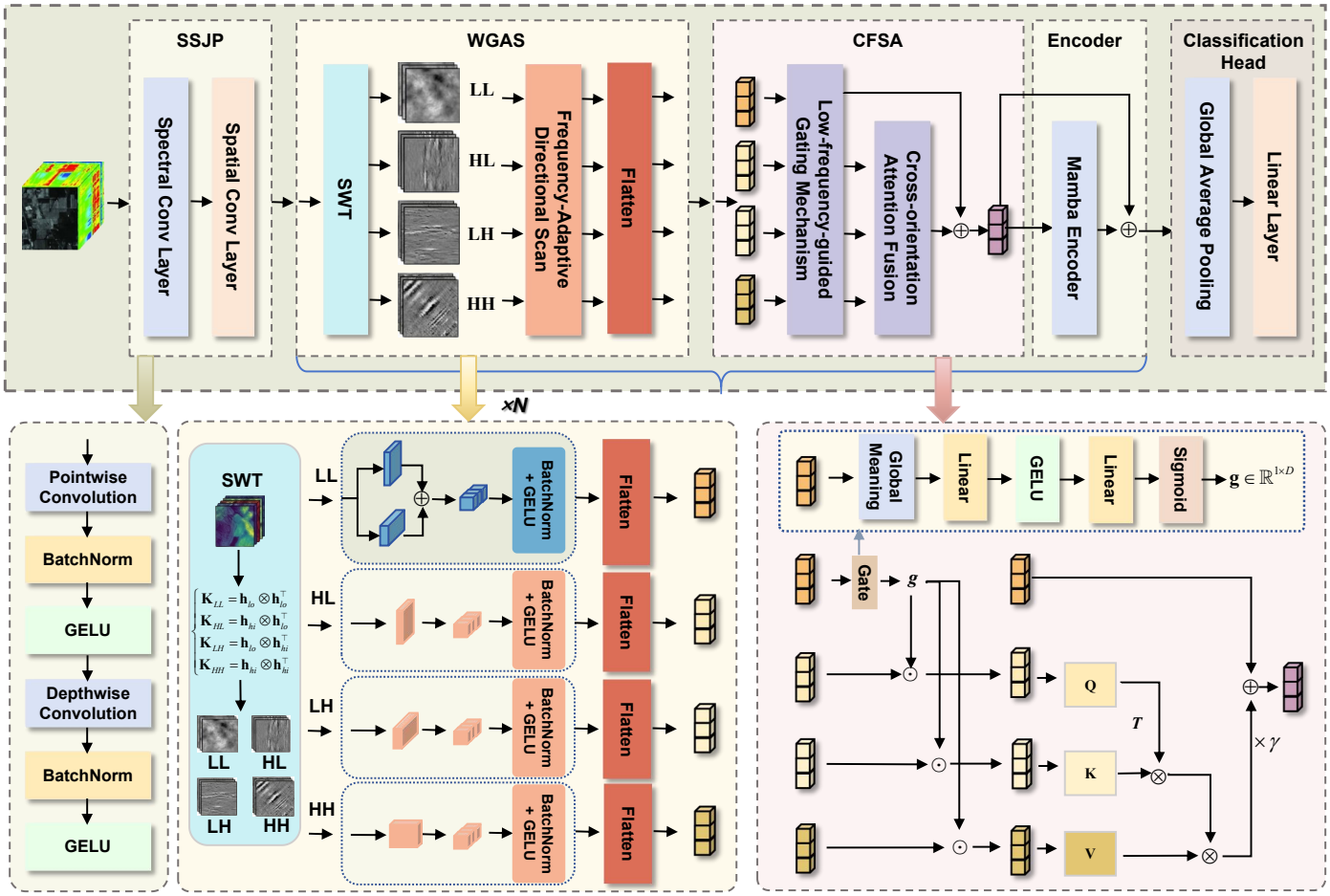


Fig. 1. The general schematic diagram of the proposed network model.

correlations that inflate the effective parameter space and heighten overfitting risks, while directly applying 3D convolution incurs prohibitive computational burdens. To address these challenges, the SSJP module performs a principled decoupling of spectral recombination and spatial context aggregation via factorized convolutions.

The input patch first passes through a Pointwise Convolution $\mathbf{W}_{1\times1} \in \mathbb{R}^{D\times C\times1\times1}$ that operates exclusively along the spectral axis. At each spatial position, this computes a learned linear combination of all C input spectral bands, projecting the redundant reflectance vector into a compact D-dimensional embedding vector. In this way, each channel synthesizes a distinct high-order spectral feature derived from the full-band correlation structure. Subsequently, a Depthwise Convolution $\mathbf{W}^{dw}_{3\times3} \in \mathbb{R}^{D\times1\times3\times3}$ with D channel groups aggregates local spatial context independently within each embedding channel, preventing spectral cross-contamination during spatial feature aggregation. The complete SSJP forward pass is formulated as:

$$\begin{aligned}\mathbf{F}_{spec} &= \delta\left(\mathcal{B}\left(\mathbf{W}_{1\times1} * \mathbf{X}\right)\right)\\ \mathbf{F}_{stem} &= \delta\left(\mathcal{B}\left(\mathbf{W}^{dw}_{3\times3} * \mathbf{F}_{spec}\right)\right), \quad \mathbf{F}_{stem} \in \mathbb{R}^{B\times D\times H\times W}\end{aligned} \tag{2}$$

where $\mathcal{B}(\cdot)$ represents Batch Normalization and δ denotes the Gaussian Error Linear Unit (GELU) activation. By factorizing spectral mixing and spatial filtering into sequential stages, SSJP circumvents the computational overhead of 3D convolution while preserving the complete joint spectral-spatial information.

C. Wavelet-Guided Anisotropic Serialization Module

After embedding, directly flattening $\mathbf{F}_{stem}$ into 1D sequences to accommodate the Mamba input format inevitably disrupts the 2D spatial topology, causing irreversible loss of geometric edge information critical for discriminating fine-grained land-cover boundaries. To resolve this incompatibility, the WGAS module projects spatial features into physically interpretable frequency-domain subspaces via channel-wise SWT and performs orientation-aware serialization tailored to each subband.

In this work, the SWT is instantiated with the Haar wavelet basis, whose filter banks can be extended to 2D for multi-orientation decomposition. Utilizing the 1D low-pass ($\mathbf{h}_{lo}$) and high-pass ($\mathbf{h}_{hi}$) filters of the Haar wavelet, four 2D orthogonal filters with specific physical orientation selectivity are constructed via tensor outer products:

$$\begin{aligned}\mathbf{K}_{LL} &= \mathbf{h}_{lo} \otimes \mathbf{h}_{lo}^{\top}, \quad \mathbf{K}_{HL} = \mathbf{h}_{hi} \otimes \mathbf{h}_{lo}^{\top}\\ \mathbf{K}_{LH} &= \mathbf{h}_{lo} \otimes \mathbf{h}_{hi}^{\top}, \quad \mathbf{K}_{HH} = \mathbf{h}_{hi} \otimes \mathbf{h}_{hi}^{\top}\end{aligned} \tag{3}$$

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The Haar wavelet is specifically chosen for its ultra-compact spatial support and distinct sensitivity to step edges. Unlike smoother, higher-order wavelets (e.g., Daubechies or Symlets) that require broader receptive fields and inevitably introduce spatial blurring, the Haar wavelet operates on a minimal 2×2 local window. This highly localized processing is particularly critical for the small spatial patches typically used in HSI classification (e.g., 7×7), as it strictly isolates horizontal, vertical, and diagonal gradients without inducing spectral aliasing across heterogeneous land-cover boundaries. Furthermore, its rectangular basis functions perfectly align with the proposed anisotropic directional scanning mechanism, facilitating a precise and computationally efficient mapping from 2D structural priors to 1D token sequences.

With this Haar wavelet as the decomposition kernel, the SWT is further deployed to process spatial features. The SWT is applied in a channel-wise manner: each of the D spectral embedding channels independently undergoes 2D wavelet decomposition with undecimated convolution to preserve the original resolution. This design ensures that the frequency decomposition respects the inter-channel spectral diversity encoded by the preceding SSJP module, rather than averaging it away:

$$\mathbf{X}_\phi = \mathbf{F}_{stem} \circledast \mathbf{K}_\phi, \quad \phi \in \{LL, HL, LH, HH\} \tag{4}$$

where $\circledast$ denotes 2D convolution with full padding. Consequently, the spatial-spectral features are decomposed into four subbands: the low-frequency global topology $\mathbf{X}_{LL}$, which captures the dominant spectral energy distribution and encodes bulk material identity; and three high-frequency subbands, including vertical ($\mathbf{X}_{HL}$), horizontal ($\mathbf{X}_{LH}$), and diagonal ($\mathbf{X}_{HH}$) gradients, which encode structurally critical edge and texture details.

Given these four frequency-domain tensors with explicit physical orientations, a frequency-adaptive directional scanning mechanism is devised to convert them into 1D sequences without destroying geometric priors. Let $\mathcal{F}_v^{3\times1}$, $\mathcal{F}_h^{1\times3}$, and $\mathcal{F}_d^{3\times3}$ denote Depthwise Convolutions with D channel groups. For $\mathbf{X}_{HL}$ and $\mathbf{X}_{LH}$, asymmetric kernels of 3×1 and 1×3 strictly align with their physical edge directions. The diagonal subband $\mathbf{X}_{HH}$ employs a 3×3 kernel to capture oblique textures. For the isotropic $\mathbf{X}_{LL}$, a cross-shaped receptive field formed by learnable weighted fusion of vertical and horizontal scans is adopted:

$$\tilde{\mathbf{X}}_\phi = \begin{cases} \mathbf{w}_1 \mathcal{F}_v^{3\times1}(\mathbf{X}_{LL}) + \mathbf{w}_2 \mathcal{F}_h^{1\times3}(\mathbf{X}_{LL}), & \phi = LL \\ \mathcal{F}_v^{3\times1}(\mathbf{X}_{HL}), & \phi = HL \\ \mathcal{F}_h^{1\times3}(\mathbf{X}_{LH}), & \phi = LH \\ \mathcal{F}_d^{3\times3}(\mathbf{X}_{HH}), & \phi = HH \end{cases} \tag{5}$$

where $\mathbf{w}_1, \mathbf{w}_2 \in \mathbb{R}^D$ are learnable channel-wise coefficients enabling per-channel adaptation of the receptive field. It is worth noting that $\mathbf{X}_{LL}$ avoids standard 3×3 convolution. The rationale behind this cross-shaped design is elaborated below: A standard dense 3×3 convolution intrinsically enforces isotropic spatial aggregation. For the already low-pass filtered $\mathbf{X}_{LL}$, such isotropic smoothing would exacerbate detail degradation and induce severe mixed-pixel aliasing at the boundaries of distinct land-cover classes. To circumvent this, the proposed cross-shaped scanning mechanism leverages spatially separable orthogonal filters. Modulated by the learnable coefficients $\mathbf{w}_1$ and $\mathbf{w}_2$, this mechanism acts as a dynamic anisotropic topological filter. This formulation explicitly empowers the network to adaptively adjust the directional bias of the receptive field. By prioritizing either horizontal or vertical spatial continuities in response to specific land-cover morphologies, it preserves the morphological skeleton of scenes without blurring fine-grained class boundaries.

Since the directional scanning relies entirely on Depthwise Convolutions, feature channels remain isolated. To restore cross-spectral communication, a 1×1 Pointwise Convolution projection layer is embedded before flattening:

$$\mathbf{S}_\phi = \mathrm{Flatten}\big(\delta\big(\mathcal{B}(\mathbf{W}_{1\times1}^{proj} * \tilde{\mathbf{X}}_\phi)\big)\big), \ \mathbf{S}_\phi \in \mathbb{R}^{N\times D} \tag{6}$$

where $N = H \times W$. The resultant 1D token sequence $\mathbf{S}_\phi$ intrinsically encodes both the frequency-specific spatial topology of subband ϕ and the cross-channel spectral correlations established by the pointwise projection, ensuring every token encapsulates a joint representation of directional spatial awareness and spectrally coherent feature abstraction.

D. Cross-Frequency Synergistic Attention Module

The inherent noise in HSIs severely degrades high-frequency textural details, whereas the low-frequency subband $\mathbf{S}_{LL}$ demonstrates natural noise robustness by encapsulating the dominant spectral energy distribution. Moreover, the spectral-spatial information distributed across the four frequency subbands is physically complementary, yet remains functionally isolated after independent directional scanning. To address these drawbacks, the proposed CFSA module synergistically integrates a low-frequency-guided gating mechanism with a cross-orientation attention fusion mechanism.

Leveraging the physical prior that $\mathbf{S}_{LL}$ captures the macroscopic spectral signature of the scene, a low-frequency-guided gating mechanism is devised. Firstly, the global spectral mean of the low-frequency sequence is computed as a semantic anchor:

$$\mathbf{p} = \frac{1}{N} \sum_{n=1}^{N} \mathbf{S}_{LL}[n, :] \in \mathbb{R}^{1\times D} \tag{7}$$

A bottleneck Multi-Layer Perceptron (MLP) with reduction ratio r transforms $\mathbf{p}$ into channel-wise gating weights $\mathbf{g} \in \mathbb{R}^{1\times D}$:

$$\mathbf{g} = \sigma\left(\mathbf{W}_{\mathrm{up}}\, \delta\left(\mathbf{W}_{\mathrm{down}}\, \mathbf{p}^\top\right)\right) \tag{8}$$

where $\sigma(\cdot)$ denotes sigmoid activation. Each high-frequency sequence is then channel-wise recalibrated:

$$\hat{\mathbf{S}}_\phi = \mathrm{LayerNorm}\left(\mathbf{S}_\phi \odot \mathbf{g}\right), \quad \phi \in \{HL, LH, HH\} \tag{9}$$

The transition from BatchNorm in the preceding SSJP module to LayerNorm in the CFSA is fundamentally dictated by the structural transformation of the feature representations. While BatchNorm effectively normalizes 2D spatial feature maps by exploiting statistics across the batch dimension, LayerNorm stabilizes the flattened 1D sequences by computing statistics independently for each token along the channel dimension.

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TABLE I
CATEGORY AND SAMPLE SETTINGS FOR THE IP, SA, HR-L, AND WHU-HC DATASETS. (TR.: TRAINING, VA.: VALIDATION, TE.: TESTING)

No.	IP				SA			
	Class Name	**Tr.**	**Va.**	**Te.**	**Class Name**	**Tr.**	**Va.**	**Te.**
1	Alfalfa	2	3	41	Brocoli weeds 1	20	20	1969
2	Corn-notill	71	72	1285	Brocoli weeds 2	37	37	3652
3	Corn-mintill	42	41	747	Fallow	20	20	1936
4	Corn	12	12	213	Fallow rough	14	14	1366
5	Grass-pasture	24	24	435	Fallow smooth	27	27	2624
6	Grass-trees	37	36	657	Stubble	39	40	3880
7	Grass-past.-mowed	2	1	25	Celery	36	36	3507
8	Hay-windrowed	24	24	430	Grapes untrain.	113	112	11046
9	Oats	1	1	18	Soil vinyard dev.	62	62	6079
10	Soybean-notill	48	49	875	Corn senesced	33	33	3212
11	Soybean-mintill	122	123	2210	Lettuce 4wk	11	10	1047
12	Soybean-clean	29	30	534	Lettuce 5wk	19	20	1888
13	Wheat	10	10	185	Lettuce 6wk	9	9	898
14	Woods	63	63	1139	Lettuce 7wk	10	11	1049
15	Bldg-Grass-Trees	20	19	347	Vinyard untrain.	73	72	7123
16	Stone-Steel-Towers	5	4	84	Vinyard trellis	18	18	1771
	Total	**512**	**512**	**9225**	**Total**	**541**	**541**	**53047**

No.	HR-L				WHU-HC			
	Class Name	**Tr.**	**Va.**	**Te.**	**Class Name**	**Tr.**	**Va.**	**Te.**
1	Dense Urban	8	9	271	Strawberry	447	448	43840
2	Mineral Extraction	2	2	63	Cowpea	228	227	22298
3	Non-Irrig. Arable	17	16	509	Soybean	103	103	10081
4	Fruit Trees	3	2	74	Sorghum	54	53	5246
5	Olive Groves	42	42	1317	Water spinach	12	12	1176
6	Broad-leaved Forest	6	7	210	Watermelon	45	46	4442
7	Coniferous Forest	15	15	470	Greens	59	59	5785
8	Mixed Forest	32	32	1008	Trees	179	180	17619
9	Dense Scler. Veg.	114	114	3565	Grass	94	95	9280
10	Sparce Scler. Veg.	84	84	2635	Red roof	105	105	10306
11	Sparcely Veg. Areas	12	12	380	Gray roof	169	169	16573
12	Rocks and Sand	14	15	458	Plastic	37	37	3605
13	Water	42	42	1309	Bare soil	91	91	8934
14	Coastal Water	14	13	424	Road	186	185	18189
15	—	—	—	—	Bright object	12	11	1113
16	—	—	—	—	Water	754	754	73893
	Total	**405**	**405**	**12693**	**Total**	**2575**	**2575**	**252380**

This normalization shift is crucial for sequence-modeling architectures. It not only guarantees statistical stability for token-to-token interactions but also prevents variance shift under the limited batch sizes typically encountered in HSI classification tasks. In this way, channels whose spectral embedding coheres with the noise-robust low-frequency features receive elevated gating coefficients, whereas noise-dominated channels are adaptively suppressed. This achieves context-aware denoising that dynamically adjusts based on the land-cover spectral composition of each input patch.

To reconstruct intact high-order spectral-spatial textures from the three orthogonal gradient subbands, a cross-orientation attention fusion mechanism is employed. The calibrated subbands are projected into distinct roles via learned linear transformations $\mathbf{W}_Q, \mathbf{W}_K \in \mathbb{R}^{D\times d_k}$ and $\mathbf{W}_V \in \mathbb{R}^{D\times d_v}$:

$$\mathbf{Q} = \hat{\mathbf{S}}_{HL}\mathbf{W}_Q,\ \mathbf{K} = \hat{\mathbf{S}}_{LH}\mathbf{W}_K,\ \mathbf{V} = \hat{\mathbf{S}}_{HH}\mathbf{W}_V \tag{10}$$

The role assignment is geometrically motivated: $\hat{\mathbf{S}}_{HL}$ serves as the query, probing for horizontally oriented structures; $\hat{\mathbf{S}}_{LH}$ provides the key, encoding complementary vertical context; and $\hat{\mathbf{S}}_{HH}$ supplies the value, contributing fine-grained textural details. Such a role assignment also carries clear mathematical implications. The horizontal query ($\mathbf{Q}$) and vertical key ($\mathbf{K}$) exhibit structural symmetry during the dot-product attention computation ($\mathbf{QK}^\top$). Consequently, their roles are strictly interchangeable, and both configurations equally model the

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TABLE II
QUANTITATIVE CLASSIFICATION RESULTS ON THE IP DATASET.

Class	CNN-based		Transformer-based		Mamba-based		
	M3D-DCNN [44]	DFFN [45]	SF [46]	SSFTT [47]	MambaHSI [38]	S^{2}Mamba [48]	**Proposed**
1	55.12 ± 21.18	57.07 ± 22.44	24.39 ± 7.07	55.28 ± 20.15	23.58 ± 22.38	**72.20 ± 10.97**	57.72 ± 16.94
2	80.90 ± 1.22	89.87 ± 4.92	67.89 ± 5.26	92.45 ± 0.64	80.65 ± 3.98	93.96 ± 2.70	**95.75 ± 2.06**
3	75.98 ± 7.39	96.65 ± 1.10	64.58 ± 6.24	94.42 ± 2.87	80.37 ± 7.48	**96.92 ± 2.69**	96.79 ± 1.42
4	71.64 ± 11.52	84.32 ± 7.46	49.01 ± 4.06	88.73 ± 1.01	79.97 ± 5.81	91.86 ± 3.26	**95.93 ± 1.81**
5	86.62 ± 4.44	92.18 ± 2.93	74.30 ± 3.99	**94.10 ± 2.18**	84.21 ± 5.76	93.84 ± 2.35	92.80 ± 3.76
6	99.09 ± 0.75	98.39 ± 0.76	91.60 ± 3.85	**99.75 ± 0.36**	95.84 ± 2.57	99.60 ± 0.28	99.54 ± 0.33
7	86.40 ± 16.32	89.60 ± 18.86	43.20 ± 24.45	90.76 ± 13.20	40.00 ± 39.60	88.80 ± 14.62	**93.33 ± 9.43**
8	99.02 ± 0.89	98.74 ± 1.48	93.95 ± 4.69	99.46 ± 0.77	97.21 ± 1.32	99.72 ± 0.56	**100.00 ± 0.00**
9	**57.78 ± 18.12**	56.67 ± 24.44	31.11 ± 8.31	55.56 ± 22.68	0.00 ± 0.00	36.67 ± 7.54	38.89 ± 23.57
10	81.03 ± 6.89	91.25 ± 2.74	75.18 ± 3.84	92.34 ± 1.12	80.72 ± 0.51	94.38 ± 0.91	**94.59 ± 3.13**
11	89.42 ± 2.59	95.48 ± 0.29	81.44 ± 2.14	96.32 ± 0.43	89.71 ± 2.20	97.16 ± 0.55	**97.51 ± 0.60**
12	83.03 ± 2.84	87.72 ± 4.19	47.87 ± 3.18	88.26 ± 0.84	73.66 ± 4.21	91.84 ± 2.73	**93.01 ± 5.15**
13	**99.78 ± 0.26**	99.14 ± 1.00	97.08 ± 1.16	99.46 ± 0.76	98.20 ± 1.02	99.68 ± 0.43	98.74 ± 0.67
OA (%)	87.12 ± 0.86	93.57 ± 1.16	76.86 ± 2.35	94.36 ± 0.42	86.18 ± 0.29	95.92 ± 0.35	**96.35 ± 0.78**
AA (%)	83.23 ± 1.78	88.50 ± 2.41	68.84 ± 2.03	88.95 ± 1.36	73.69 ± 1.46	90.11 ± 1.43	**90.09 ± 2.15**
$\kappa\times100$	85.27 ± 0.96	92.67 ± 1.32	73.59 ± 2.66	93.56 ± 0.48	84.19 ± 0.30	95.34 ± 0.39	**95.83 ± 0.89**

intersection geometries of orthogonal edges. Conversely, the $\hat{\mathbf{S}}_{HH}$ must exclusively serve as the value ($\mathbf{V}$). Rather than computing structural correlations, $\mathbf{V}$ provides the substantive fine-grained textural representations that are subsequently aggregated based on the orthogonal attention weights, making its role non-interchangeable with $\mathbf{Q}$ or $\mathbf{K}$. The cross-orientation attention fusion mechanism is computed as:

$$\mathbf{A}_{\text{high}} = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}\mathbf{W}_{\text{out}} \tag{11}$$

where $\mathbf{W}_{\text{out}} \in \mathbb{R}^{d_v \times D}$. This operation leverages the orthogonal interaction of horizontal and vertical gradients to dynamically aggregate complex diagonal textures, synthesizing a structurally enriched high-frequency representation.

The derived high-frequency attention tensor is integrated with the low-frequency backbone via a zero-initialized residual connection:

$$\mathbf{S}_{\text{fused}} = \text{LayerNorm}\left(\mathbf{S}_{LL} + \gamma\mathbf{A}_{\text{high}}\right) \tag{12}$$

where $\gamma \in \mathbb{R}$ is a learnable scalar initialized to zero [43], ensuring stable spectral initialization at training onset while progressively incorporating structurally informative high-frequency details as γ increases.

While the preceding WGAS and CFSA modules excel at extracting direction-aware, noise-suppressed, and physically interpretable local features, their operations are fundamentally constrained by localized receptive fields. In isolation, they cannot capture the macroscopic spatial context or the long-range inter-pixel dependencies across the entire HSI patch. Therefore, the Mamba Encoder is integrated at the end of the pipeline to establish a complementary "local-to-global" learning paradigm. Under this architecture, WGAS and CFSA inject robust 2D geometric priors into 1D sequences. Subsequently, the Mamba Encoder leverages its linear-complexity SSM to propagate these refined local tokens across the entire sequence. This synergistic stacking ensures that the final representation encapsulates both fine-grained local boundaries and comprehensive global spatial-spectral contexts. So, the fused sequence $\mathbf{S}_{\text{fused}}$ is sent to the Mamba Encoder. The selective SSM discretizes a continuous SSM with input-dependent parameters, enabling content-aware selective information propagation:

$$\mathbf{h}_t = \bar{\mathbf{A}}_t\mathbf{h}_{t-1} + \bar{\mathbf{B}}_t\mathbf{x}_t, \quad \mathbf{y}_t = \bar{\mathbf{C}}_t\mathbf{h}_t \tag{13}$$

where $\mathbf{x}_t = \mathbf{S}_{\text{fused}}[t,:]$, and the discretized parameters $\bar{\mathbf{A}}_t, \bar{\mathbf{B}}_t, \bar{\mathbf{C}}_t$ are dynamically generated as functions of $\mathbf{x}_t$. Because the input tokens have undergone SSJP, WGAS, and CFSA module, each $\mathbf{x}_t$ intrinsically encodes joint spectral-spatial-frequency priors. The recurrent hidden state $\mathbf{h}_t$ efficiently propagates these priors across the full sequence, establishing global long-range dependencies at strictly linear $\mathcal{O}(N)$ complexity. Finally, the output sequence undergoes global average pooling and a linear classification head to produce the class probability distribution.

III. EXPERIMENTS

A. Datasets

To illustrate the classification ability of the proposed CFAS-Mamba, four benchmark HSI datasets are used for comprehensive evaluation, including Indian Pines (IP), Salinas (SA), HyRANK-Loukia (HR-L), and Wuhan UAV-borne HSI-HanChuan (WHU-HC). The detailed description is as follows.

1) Indian Pines (IP): The IP dataset covers a spectral range of 0.4–2.5 μm with 220 bands. Following the exclusion of 20 noisy bands, the remaining 200 channels were retained for analysis. This dataset features a spatial dimension of 145 × 145 pixels and comprises 10,249 labeled samples distributed across 16 distinct land-cover classes.

2) Salinas (SA): The SA dataset comprises 224 spectral bands spanning the 400 to 2500 nm range. For experimental purposes, 20 noisy bands were discarded, leaving 204 valid bands for analysis. The image features spatial dimensions of 512 × 217 pixels and contains a total of 54,129 annotated pixels categorized into 16 distinct classes.

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TABLE III
QUANTITATIVE CLASSIFICATION RESULTS ON THE SA DATASET.

Class	CNN-based		Transformer-based		Mamba-based		
	M3D-DCNN [44]	DFFN [45]	SF [46]	SSFTT [47]	MambaHSI [38]	S^{2}Mamba [48]	**Proposed**
1	98.91 ± 1.19	97.14 ± 1.85	95.71 ± 1.24	**100.00 ± 0.00**	99.58 ± 0.46	98.09 ± 1.95	99.56 ± 0.27
2	**99.98 ± 0.03**	99.80 ± 0.14	99.48 ± 0.26	99.66 ± 0.52	99.17 ± 0.47	98.73 ± 1.43	**99.99 ± 0.01**
3	95.76 ± 2.50	95.34 ± 2.32	95.19 ± 1.28	96.73 ± 1.71	93.58 ± 1.71	98.23 ± 2.11	**98.55 ± 0.75**
4	97.47 ± 1.57	97.09 ± 1.21	94.55 ± 2.30	99.22 ± 0.61	97.34 ± 0.66	**99.51 ± 0.14**	99.34 ± 0.36
5	98.19 ± 1.02	95.58 ± 3.07	92.20 ± 4.00	96.59 ± 2.68	96.85 ± 2.28	98.41 ± 0.31	**98.98 ± 0.64**
6	99.75 ± 0.23	99.79 ± 0.37	99.42 ± 0.58	99.69 ± 0.37	99.96 ± 0.06	**100.00 ± 0.00**	99.99 ± 0.01
7	99.83 ± 0.06	99.53 ± 0.22	98.25 ± 0.70	99.70 ± 0.21	99.98 ± 0.03	**100.00 ± 0.00**	99.93 ± 0.05
8	86.55 ± 2.55	90.19 ± 2.92	84.97 ± 2.03	87.59 ± 1.46	86.93 ± 2.76	89.02 ± 0.23	**92.79 ± 3.06**
9	98.87 ± 1.64	97.67 ± 1.75	96.76 ± 1.44	98.91 ± 1.22	99.56 ± 0.47	99.34 ± 0.73	**99.98 ± 0.02**
10	92.96 ± 1.55	91.96 ± 0.83	89.64 ± 2.80	94.51 ± 1.71	**96.94 ± 1.09**	96.03 ± 2.86	95.25 ± 3.74
11	97.42 ± 1.98	96.37 ± 3.19	90.05 ± 1.77	96.70 ± 2.25	92.90 ± 1.68	98.95 ± 1.15	**98.98 ± 0.32**
12	99.34 ± 0.59	99.74 ± 0.35	98.47 ± 1.38	99.88 ± 0.07	98.02 ± 0.15	**99.98 ± 0.02**	**99.98 ± 0.02**
13	97.35 ± 3.16	98.46 ± 2.69	92.61 ± 7.78	97.84 ± 2.66	**99.70 ± 0.28**	98.85 ± 1.01	93.36 ± 6.88
14	96.34 ± 2.01	97.03 ± 1.71	96.57 ± 2.67	91.80 ± 4.85	**99.08 ± 0.67**	98.79 ± 0.09	97.08 ± 1.60
15	83.67 ± 3.78	85.34 ± 5.37	79.07 ± 4.88	81.42 ± 3.77	82.53 ± 7.17	**86.13 ± 1.70**	85.98 ± 3.82
16	94.65 ± 3.13	95.87 ± 2.88	92.72 ± 4.21	**97.09 ± 1.58**	93.58 ± 4.13	93.00 ± 3.85	96.14 ± 0.95
OA (%)	93.70 ± 0.35	94.30 ± 0.43	91.32 ± 0.78	93.73 ± 0.39	93.71 ± 0.79	94.92 ± 0.24	**95.87 ± 0.50**
AA (%)	96.06 ± 0.29	96.06 ± 0.29	93.48 ± 0.73	96.08 ± 0.47	95.98 ± 0.79	97.07 ± 0.17	**97.24 ± 0.35**
$\kappa\times100$	92.99 ± 0.39	93.65 ± 0.48	90.34 ± 0.88	93.01 ± 0.44	93.00 ± 0.88	94.35 ± 0.27	**95.39 ± 0.55**

TABLE IV
QUANTITATIVE CLASSIFICATION RESULTS ON THE HR-L DATASET.

Class	CNN-based		Transformer-based		Mamba-based		
	M3D-DCNN [44]	DFFN [45]	SF [46]	SSFTT [47]	MambaHSI [38]	S^{2}Mamba [48]	**Proposed**
1	54.39 ± 8.49	37.71 ± 17.86	47.75 ± 9.05	69.99 ± 2.56	45.39 ± 5.93	66.17 ± 9.74	**72.69 ± 14.55**
2	**90.79 ± 12.55**	79.68 ± 16.35	98.41 ± 3.17	83.07 ± 22.83	70.37 ± 12.59	77.25 ± 16.41	82.01 ± 9.19
3	**82.28 ± 3.25**	80.16 ± 4.35	66.44 ± 6.55	74.72 ± 3.46	74.00 ± 3.96	76.69 ± 1.62	82.25 ± 1.30
4	30.54 ± 11.36	4.86 ± 7.33	1.89 ± 2.36	**54.95 ± 3.55**	9.01 ± 8.43	9.46 ± 7.24	33.33 ± 17.67
5	85.92 ± 3.12	88.93 ± 3.48	70.90 ± 4.22	83.30 ± 1.87	85.95 ± 3.74	86.81 ± 2.98	**88.86 ± 6.33**
6	48.67 ± 10.67	27.52 ± 6.41	13.05 ± 7.07	**48.73 ± 9.56**	21.90 ± 11.74	29.68 ± 7.06	46.51 ± 7.35
7	**64.55 ± 8.41**	46.98 ± 9.40	42.21 ± 5.77	64.26 ± 6.66	48.72 ± 7.34	57.66 ± 9.46	57.45 ± 2.80
8	68.61 ± 8.23	61.75 ± 7.19	62.28 ± 6.11	70.14 ± 9.45	68.06 ± 4.44	66.73 ± 13.23	**71.73 ± 3.55**
9	78.91 ± 3.89	82.12 ± 2.17	72.07 ± 4.82	78.88 ± 1.02	83.01 ± 3.87	**84.25 ± 5.93**	83.97 ± 3.15
10	**81.98 ± 3.66**	78.68 ± 4.04	71.43 ± 5.95	79.95 ± 1.98	73.03 ± 3.72	78.47 ± 1.22	79.61 ± 4.59
11	60.42 ± 6.80	56.89 ± 5.59	57.11 ± 5.50	**69.74 ± 5.48**	53.95 ± 8.67	65.61 ± 2.37	70.09 ± 5.72
12	92.58 ± 4.03	90.13 ± 2.82	91.09 ± 2.89	93.89 ± 1.54	**94.47 ± 0.54**	94.32 ± 1.11	91.05 ± 2.59
13	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
14	**100.00 ± 0.00**	**100.00 ± 0.00**	99.95 ± 0.09	98.90 ± 0.11	**100.00 ± 0.00**	99.84 ± 0.22	99.53 ± 0.51
OA (%)	80.63 ± 1.15	78.78 ± 0.87	71.98 ± 0.98	80.46 ± 0.09	77.98 ± 0.52	80.83 ± 0.47	**82.40 ± 0.85**
AA (%)	74.26 ± 1.15	66.82 ± 2.08	63.90 ± 0.47	**76.46 ± 1.80**	66.28 ± 3.69	70.93 ± 2.06	75.65 ± 0.98
$\kappa\times100$	76.96 ± 1.25	74.54 ± 1.06	66.54 ± 1.06	76.82 ± 0.17	73.59 ± 0.69	77.06 ± 0.54	**78.96 ± 1.11**

3) HyRANK-Loukia (HR-L): The HR-L dataset consists of hyperspectral imagery with 176 spectral channels and a spatial resolution of 249 × 945 pixels. It provides a total of 13,503 manually annotated pixels, which are categorized into 14 distinct object classes.

4) WHU-Hi-HanChuan (WHU-HC): The WHU-HC dataset is characterized by 270 spectral bands spanning the wavelength range of 400 to 1000 nm, with a spatial resolution of 0.109 m. The image dimensions are 1217 × 303 pixels, yielding a total of 257,530 labeled pixels categorized into 16 distinct classes.

The labeled samples were randomly partitioned into training, validation, and testing subsets with dataset-specific ratios. Specifically, the IP dataset employs a 5%, 5%, and 90% split. In contrast, a more challenging limited-sample setting is adopted for the SA and WHU-HC datasets, allocating only 1% each for training and validation, with the remaining 98% reserved for testing. For the HR-L dataset, the partition is set to 3%, 3%, and 94%, respectively. Detailed distributions of these splits are tabulated in Tables I–IV.

Furthermore, to mitigate the potential overfitting caused by sample scarcity, the training subset was augmented using geometric transformations. These strategies include vertical and horizontal flipping, as well as rotations at 90°, 180°, and 270° intervals.

B. Experimental Settings

1) Evaluation Metrics: In order to objectively evaluate and quantitatively analyze the performance of various neural networks in HSI classification, three indicators are used to evaluate all methods, including Overall Accuracy (OA), Average Accuracy (AA) and $Kappa$ coefficient.

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TABLE V
CLASSIFICATION RESULTS OF DIFFERENT CLASSIFICATION METHODS ON THE WHU-HC DATASET WITH 1% TRAINING SAMPLES.

Class	CNN-based		Transformer-based		Mamba-based		
	M3D-DCNN [44]	DFFN [45]	SF [46]	SSFTT [47]	MambaHSI [38]	S^2Mamba [48]	**Proposed**
1	96.67 ± 0.36	97.29 ± 0.68	96.30 ± 0.78	95.67 ± 0.75	95.59 ± 2.35	58.54 ± 13.94	**98.29 ± 0.23**
2	91.07 ± 2.40	95.07 ± 1.55	84.95 ± 2.43	90.06 ± 1.74	90.88 ± 2.85	92.61 ± 2.73	**97.04 ± 0.10**
3	90.51 ± 1.92	95.42 ± 1.26	91.40 ± 1.62	85.19 ± 1.94	89.16 ± 2.61	96.03 ± 1.42	**97.02 ± 0.55**
4	96.23 ± 1.21	96.54 ± 1.07	95.33 ± 0.67	91.68 ± 4.01	95.80 ± 1.03	95.15 ± 2.88	**96.61 ± 0.62**
5	62.04 ± 11.86	86.80 ± 7.24	66.19 ± 9.42	53.06 ± 10.28	64.71 ± 8.89	**95.71 ± 1.97**	85.94 ± 4.87
6	53.81 ± 5.83	64.98 ± 7.25	41.58 ± 3.59	52.87 ± 4.43	54.01 ± 2.68	**99.44 ± 0.31**	76.68 ± 4.41
7	84.12 ± 5.07	89.38 ± 2.63	80.03 ± 2.37	78.65 ± 1.81	84.49 ± 4.21	76.00 ± 33.94	**92.87 ± 2.15**
8	82.53 ± 3.82	88.61 ± 0.11	77.98 ± 2.19	82.42 ± 0.45	88.41 ± 1.29	**99.69 ± 0.44**	91.95 ± 0.64
9	83.85 ± 2.49	91.23 ± 1.49	71.73 ± 3.68	75.29 ± 3.89	85.58 ± 3.68	40.74 ± 34.35	**95.13 ± 1.93**
10	95.77 ± 0.71	**98.47 ± 0.37**	96.65 ± 0.44	94.43 ± 2.22	95.91 ± 0.23	93.79 ± 4.10	97.90 ± 0.81
11	95.22 ± 1.00	95.81 ± 1.53	95.07 ± 1.35	95.36 ± 1.10	95.56 ± 0.32	**97.06 ± 0.87**	96.20 ± 1.81
12	70.37 ± 6.07	78.80 ± 7.46	55.23 ± 3.87	57.64 ± 3.42	76.24 ± 2.40	**92.57 ± 1.68**	88.21 ± 3.82
13	70.85 ± 2.04	79.95 ± 2.16	67.34 ± 2.91	60.58 ± 3.40	77.17 ± 4.02	**99.46 ± 0.44**	86.43 ± 1.30
14	89.72 ± 1.90	91.93 ± 1.88	87.90 ± 1.41	85.04 ± 1.21	88.49 ± 1.78	**98.95 ± 1.12**	93.25 ± 0.96
15	81.53 ± 3.60	79.30 ± 6.66	62.26 ± 12.18	81.67 ± 9.05	76.34 ± 5.29	**90.49 ± 3.48**	81.55 ± 12.47
16	99.30 ± 0.26	99.30 ± 0.22	96.90 ± 2.21	99.17 ± 0.27	98.95 ± 0.25	94.05 ± 5.05	**99.73 ± 0.13**
OA (%)	92.04 ± 0.41	94.57 ± 0.70	89.92 ± 0.30	90.00 ± 0.83	92.39 ± 0.27	95.64 ± 0.87	**96.24 ± 0.49**
AA (%)	83.97 ± 0.93	89.30 ± 1.91	79.32 ± 0.78	79.92 ± 1.65	84.83 ± 0.59	88.77 ± 0.42	**92.18 ± 1.82**
$\kappa\times100$	90.67 ± 0.48	93.65 ± 0.81	88.07 ± 0.35	88.27 ± 0.98	91.10 ± 0.31	95.02 ± 0.99	**95.60 ± 0.57**

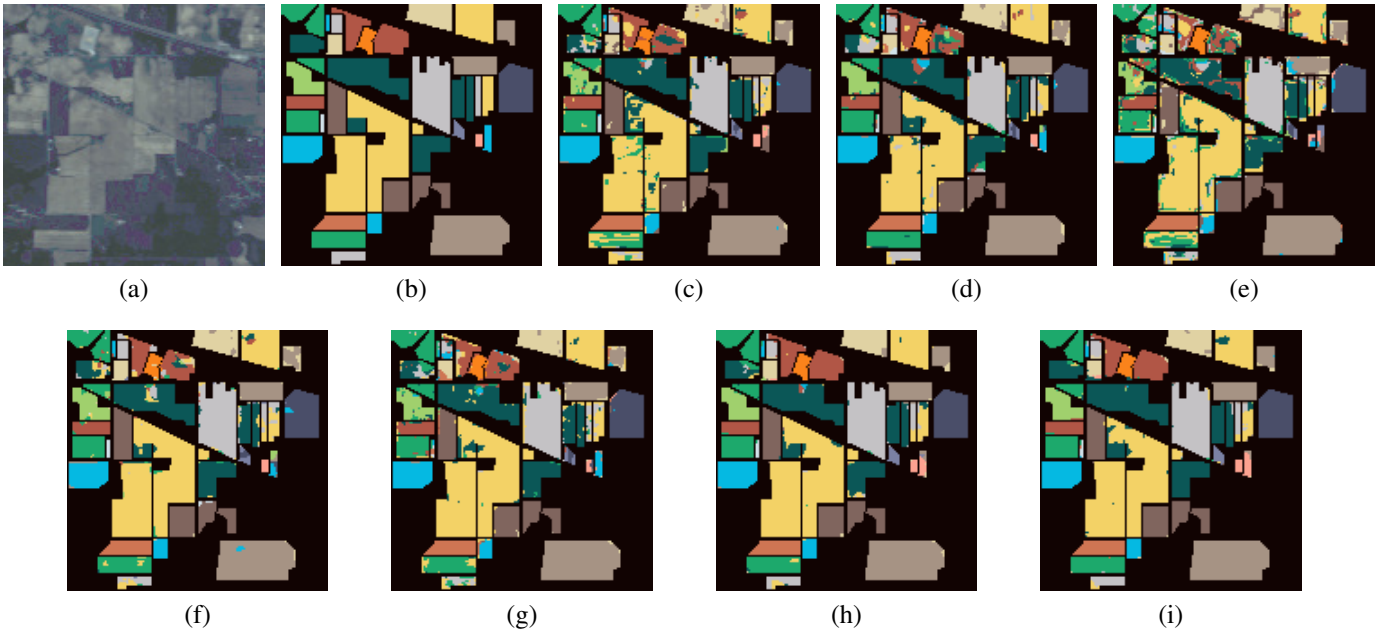


Fig. 2. Qualitative visualization of the classification maps for the IP dataset. (a) False-Color Image. (b) Ground Truth. (c) M3D-DCNN. (d) DFFN. (e) SF. (f) SSFTT. (g) MambaHSI. (h) S^2Mamba. (i) CFAS-Mamba.

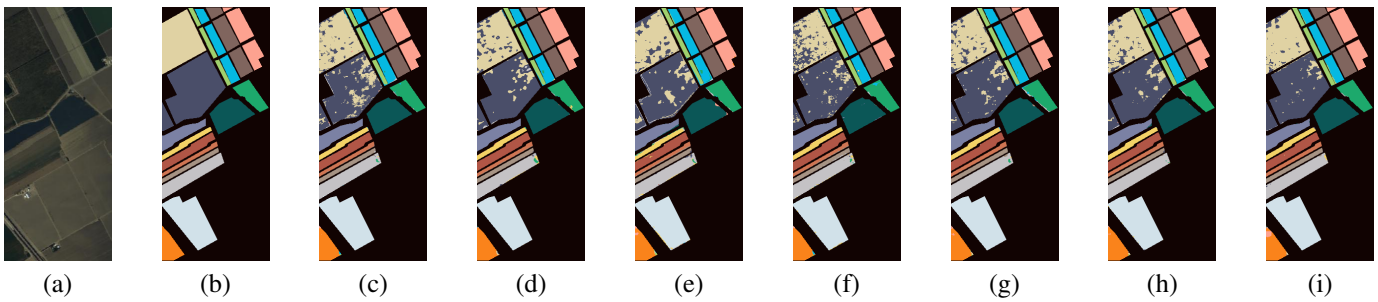


Fig. 3. Qualitative visualization of the classification maps for the SA dataset. (a) False-Color Image. (b) Ground Truth. (c) M3D-DCNN. (d) DFFN. (e) SF. (f) SSFTT. (g) MambaHSI. (h) S^2Mamba. (i) CFAS-Mamba.

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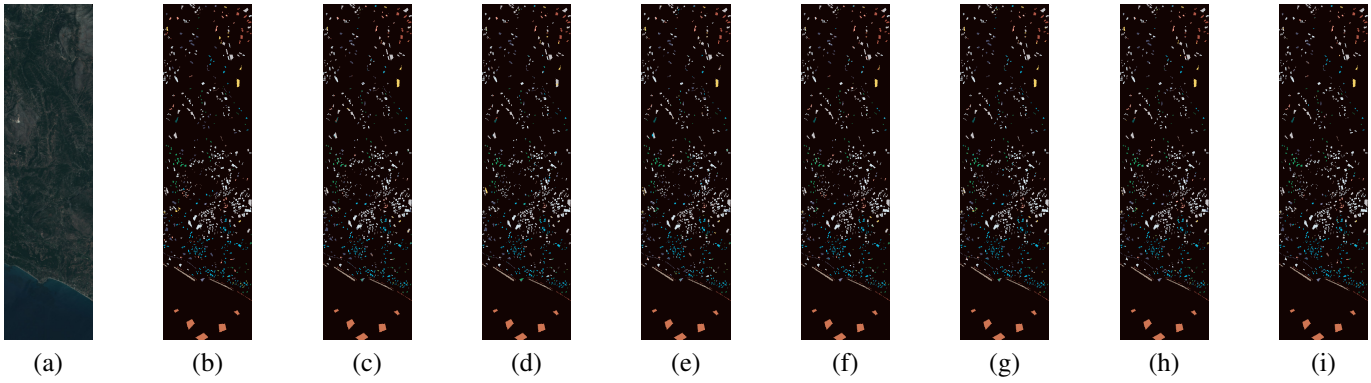


Fig. 4. Qualitative visualization of the classification maps for the HR-L dataset. (a) False-Color Image. (b) Ground Truth. (c) M3D-DCNN. (d) DFFN. (e) SF. (f) SSFTT. (g) MambaHSI. (h) S^2Mamba. (i) CFAS-Mamba.

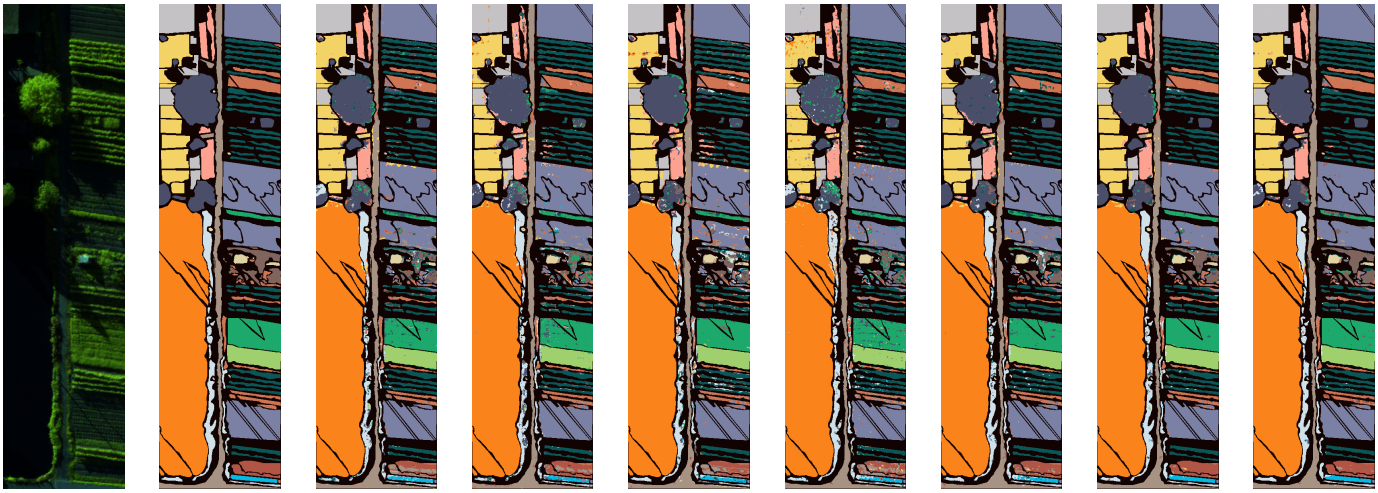


Fig. 5. Qualitative visualization of the classification maps for the WHU-HC dataset. (a) False-Color Image. (b) Ground Truth. (c) M3D-DCNN. (d) DFFN. (e) SF. (f) SSFTT. (g) MambaHSI. (h) S^2Mamba. (i) CFAS-Mamba.

2) Implementation Details: All comparative experiments were executed with a uniform batch size of 64 and a duration of 300 epochs to guarantee consistency. The optimization protocols for the competing methods followed the settings specified in their respective studies. In contrast, our proposed framework employed Stochastic Gradient Descent (SGD) for parameter updating, utilizing a constant learning rate of 0.001, a weight decay of 0.0001, and a momentum factor of 0.9. All computations were conducted on an NVIDIA GeForce RTX 4090 GPU equipped with 24GB memory.

C. Experimental Comparison With Competitive Approaches

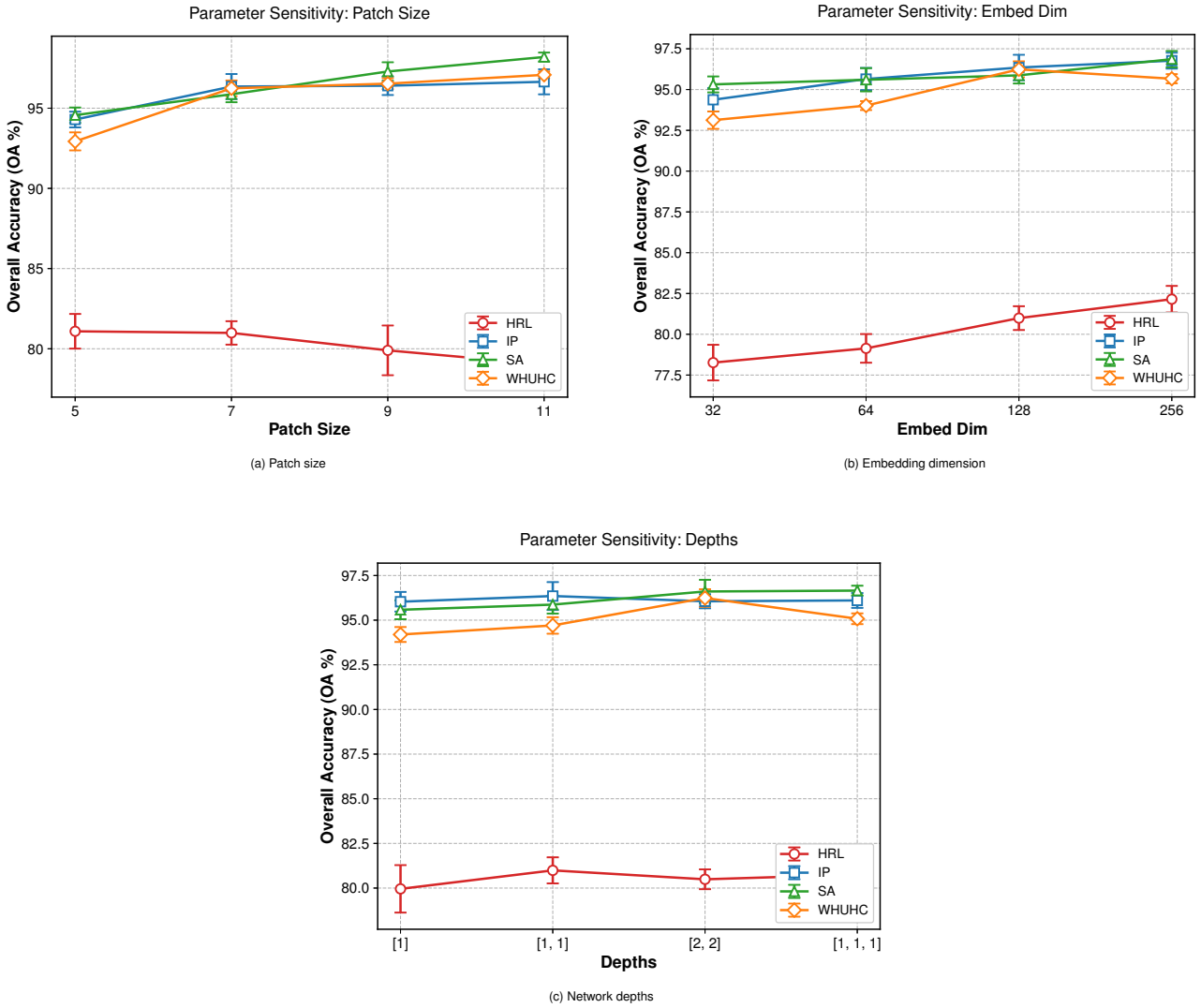
To comprehensively evaluate the performance of the proposed CFAS-Mamba, several SOTA HSI classification methods are selected for comparative analysis. These benchmarks span different architectural paradigms, including Convolutional Neural Networks (CNNs), Transformer-based models, and Mamba-based models. Specifically: CNN-based methods include M3D-DCNN [44], which employs multi-scale 3D convolutions, and DFFN [45], which focuses on deep feature fusion; Transformer-based methods consist of SF [46] and SSFTT [47], which leverage various attention mechanisms for spectral-spatial tokenization; Mamba-based methods include MambaHSI [38] and S^2Mamba [48], which excel in capturing long-range dependencies with SSMs.

The experimental results comprehensively demonstrate the SOTA performance of CFAS-Mamba across four benchmark datasets. As shown in Tables II to V, compared with CNNs and Transformers, the CFAS-Mamba significantly outperforms the competing models on the IP and SA datasets, achieving remarkable OA scores of 96.35% and 95.87%, respectively. A horizontal comparison reveals that the Mamba-based models exhibits extremely strong overall competitiveness, proving the immense potential of SSM in processing high-dimensional continuous spectral data. However, early models like MambaHSI inevitably destroy spatial local continuity when flattening image patches into sequences. The proposed CFAS-Mamba effectively mitigates this structural limitation by WGAS and CFSA module. It not only secures high classification accuracy within homogeneous regions but also exhibits robust discriminative capabilities along easily confused transitional categories.

Class-imbalance remains a critical bottleneck in HSI classification; however, the proposed CFAS-Mamba mitigates this

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(a) Patch size

(b) Embedding dimension

(c) Network depths

Fig. 6. Sensitivity analysis of hyperparameters on four datasets: (a) patch size, (b) embedding dimension, and (c) network depths.

limitation by fostering robust class equilibrium. As evidenced by the WHU-HC dataset under a constrained 1% training configuration (Table V), the baseline S^{2}Mamba exhibits a competitive 95.64% OA but undergoes a severe prediction collapse on the minority class (Class 9, Grass), yielding a mere 40.74% accuracy. Conversely, CFAS-Mamba consistently surpasses all counterparts, delivering an OA of 96.24%, a κ of 95.60%, and a well-balanced AA of 92.18%, while notably reclaiming a 95.13% accuracy on the critical "Grass" category. This performance profile demonstrates that the low-frequency-guided gating mechanism effectively suppresses the feature representation clipping of rare classes by dominant categories. Moreover, on the highly heterogeneous and fragmented HR-L dataset (Table IV), where conventional Transformers are bounded by a 71.98% OA, the proposed model advances the OA to 82.40%. Leveraging frequency-domain scanning and cross-orientation attention fusion for noise-resistant feature extraction, CFAS-Mamba maintains stable performance across highly dispersed and spectrally mixed landscapes.

D. Parameter Analysis

To establish the optimal hyperparameter configuration for CFAS-Mamba, a systematic sensitivity analysis of spatial patch size, embedding dimension, and network depth are conducted. Fig. 6 reports the OA across all four benchmark datasets under varying parameter settings.

1) Effect of Spatial Patch Size: The spatial patch size determines the receptive field from which spectral-spatial features are extracted. As illustrated in Fig. 6(a), the OA on the IP, SA, and WHU-HC datasets exhibits a upward trend as the patch size increases from 5 to 11. The most pronounced gain occurs between 5 and 7, after which the improvement becomes gradual. This behavior is consistent with the intrinsic spatial homogeneity of these scenes, where an expanded receptive field captures richer contextual cues and reinforces intra-class consistency.

In stark contrast, the HR-L dataset displays an opposite trend: the OA declines monotonically as the patch size enlarges. This degradation reflects the highly fragmented land-cover distribution characteristic of HR-L, where larger patches

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inevitably introduce boundary-mixed pixels that dilute edge-discriminative information. The increasing proportion of heterogeneous pixels within the receptive field overwhelms the benefit of additional spatial context, directly impairing classification fidelity at land-cover boundaries. Based on the above observations, a unified patch size of 7 is selected for all experiments. This size acts as a universal optimal threshold: it is large enough to capture the critical spatial contextual cues required by homogeneous scenes (IP, SA, WHUHC), yet moderate enough to prevent the inclusion of noisy, boundary-mixed pixels in highly fragmented scenes (HR-L).

2) Effect of Embedding Dimension: The embedding dimension *embed_dim* governs the channel capacity of the SSM and the cross-orientation attention fusion. Fig. 6(b) depicts the OA response over the candidate set $\{32, 64, 128, 256\}$. On the IP, SA, and HR-L datasets, the OA exhibits a monotonic improvement as the embedding dimension increases. The gain from 32 to 128 is substantial, while the further increment to 256 yields only marginal additional benefit. This indicates that expanded channel capacity consistently enhances the model's representational power, albeit with diminishing returns beyond a certain threshold.

On the WHUHC dataset, a slightly different pattern emerges: the OA peaks at 128 and undergoes a mild recession at 256. This suggests that under the limited training regime of WHUHC, an excessively high-dimensional feature space incurs mild overfitting, where the model begins to encode dataset-specific high-frequency noise at the expense of transferable representations. To balance representational capacity against generalization across all four scenes, *embed_dim* $=$ 128 is selected as the unified configuration, achieving near-optimal OA on every dataset without the overparameterization risk associated with higher dimensions.

3) Effect of Network Depth: In CFAS-Mamba, network depth is parameterized as a list *depths*: the list length specifies the number of hierarchical stages (each comprising WGAS, CFAS, and Mamba Encoder), while each element defines the number of cascaded Mamba Encoders within the corresponding stage. Four representative configurations are evaluated: **[1]** (single-stage), **[1,1]** (dual-stage), **[2,2]** (deep intra-stage stacking), and **[1,1,1]** (triple-stage).

As shown in Fig. 6(c), the single-stage architecture **[1]** consistently yields the lowest OA across all datasets, confirming that a single pass through the frequency-domain pipeline is insufficient for extracting multi-scale representations. Beyond this baseline, the optimal depth configuration varies with scene characteristics. On IP and HR-L, **[1,1]** achieves the highest OA, with deeper configurations **[2,2]** and **[1,1,1]** delivering comparable or slightly degraded performance. This suggests that for these datasets, two hierarchical stages strike an optimal balance between feature abstraction and spatial detail retention. On SA, a monotonic improvement with architectural depth is observed: **[1,1,1]** attains the best OA, followed closely by **[2,2]**, indicating that the large-scale homogeneous agricultural landscape benefits from deeper hierarchical decomposition. On WHU-HC, **[2,2]** is optimal, demonstrating that intra-stage depth, which stacks Mamba blocks within fewer stages, captures the fine-grained spectral-spatial patterns of this complex scene more effectively than inter-stage depth.

These dataset-specific preferences notwithstanding, **[1,1]** is adopted as the unified depth configuration. It achieves the highest OA on two datasets (IP and HR-L), performs competitively on SA, and provides a favorable trade-off between representational depth and computational efficiency across all evaluated scenes.

E. Ablation Experiments

To evaluate the individual contribution and synergistic effects of each component within CFAS-Mamba, a systematic ablation study is conducted. Following standard literature conventions, the ablation analysis is dedicated to the IP dataset due to its severe spectral-spatial aliasing, strong sensor noise, and highly unbalanced long-tail category distribution. The architectural components are categorized into three functional groups: (i) the WGAS module, (ii) the CFSA module, and (iii) the Mamba Encoder. Table VI reports the quantitative classification performance under various ablated configurations.

1) Contribution of the WGAS Module: The WGAS module decomposes the spectral-spatial representation into physically meaningful frequency subbands and serializes them in a topology-preserving manner. Removing the Wavelet Decomposition framework (*w/o Wavelet*) disables the entire downstream frequency-domain pipeline, causing the OA and AA to decrease to 93.56% and 86.35%, respectively. This sharp decline demonstrates that frequency-domain decoupling is a foundational prerequisite of CFAS-Mamba, without which the model degenerates into a naive sequence processor ill-equipped to handle complex spatial-spectral aliasing.

Within the WGAS pipeline, the frequency-adaptive directional scanning mechanism proves critical. Replacing the anisotropic directional convolutions (1×3 and 3×1) with conventional isotropic 3×3 convolutions (*w/o Dir. Scan*) reduces the OA by 2.60 percentage points to 93.75% and the AA to 87.81%. This degradation validates that horizontal and vertical high-frequency subbands encode structurally directional edge information that isotropic kernels inherently blur. Additionally, removing the 1×1 pointwise convolution (*w/o* 1×1 *Conv*) reduces the OA to 94.93% and the AA to 88.61%, confirming that this operation is indispensable for re-establishing cross-channel spectral correlations after independent directional scanning.

2) Contribution of the CFSA Module: The CFSA module operates on the frequency-domain tokens to facilitate cross-frequency synergistic interactions. Its importance is most starkly revealed by ablating the attention fusion mechanism (*w/o Attn. Fusion*), where the learned cross-orientation attention fusion is replaced with naive element-wise summation. This substitution triggers the most severe performance collapse across all single-component ablations, causing the OA to plunge to 93.02% and the AA to drop to 85.72%. This decline indicates strong semantic heterogeneity among the directionally scanned subbands, meaning that rigid linear superposition destroys the high-frequency textural details that our cross-orientation attention fusion mechanism otherwise adaptively reconstructs.

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TABLE VI
ABLATION STUDY OF CFAS-MAMBA ON THE IP DATASET. THE COMPONENTS ARE GROUPED BY ARCHITECTURAL MODULE ALONG THE INFERENCE PIPELINE. WAVE.: WAVELET DECOMPOSITION; DIR.: FREQUENCY-ADAPTIVE DIRECTIONAL SCANNING MECHANISM; CONV.: 1×1 POINTWISE CONV ; GAT.: LOW-FREQUENCY-GUIDED GATING MECHANISM; ATTN.: CROSS-ORIENTATION ATTENTION FUSION MECHANISM; MAM.: MAMBA ENCODER. "–" DENOTES COMPONENTS DISABLED BY CASCADE DEPENDENCY. THE BEST RESULTS ARE IN **BOLD**.

Configuration	WGAS			CFSA		Backbone	Metrics		
	Wave.	Dir.	Mix	Gat.	Attn.	Mam.	OA (%)	AA (%)	$\kappa \times 100$
w/o Wavelet	✗	–	–	–	–	✓	93.56	86.35	92.65
w/o Dir. Scan	✓	✗	✓	✓	✓	✓	93.75	87.81	92.86
w/o 1×1 Conv	✓	✓	✗	✓	✓	✓	94.93	88.61	94.22
w/o Gating	✓	✓	✓	✗	✓	✓	94.39	88.36	93.60
w/o Attn. Fusion	✓	✓	✓	✓	✗	✓	93.02	85.72	92.02
w/o Mamba (Trans.)	✓	✓	✓	✓	✓	✗	94.23	88.00	93.41
Full CFAS-Mamba	✓	✓	✓	✓	✓	✓	**96.35**	**90.09**	**95.83**

The low-frequency-guided gating mechanism also proves consequential. Its removal (*w/o Gating*) reduces the OA to 94.39% and the AA to 88.36%. This confirms that unconstrained high-frequency activations introduce pervasive spatial noise, and that our gating mechanism effectively suppresses such interference without sacrificing geometric boundaries.

3) Contribution of the Sequence Modeling Backbone: To evaluate the efficacy of the Mamba Encoder, we replace it with a standard multi-head self-attention Transformer while retaining all frequency-domain components (*w/o Mamba (Trans.)*). This substitution yields an OA of 94.23% and an AA of 88.00%, representing declines of 2.12 and 2.09 percentage points relative to the full model. This performance gap corroborates the advantages of Mamba's selective state space mechanism. Its linear recurrence with input-dependent parameterization captures long-range contextual dependencies more effectively than global self-attention, which remains susceptible to information scattering when operating on prolonged, high-dimensional spectral tokens.

4) Synergistic Integration: Synthesizing the ablation results, a clear hierarchical dependency emerges among the modules. The cross-orientation attention fusion serves as the ultimate semantic bridge, while the wavelet decomposition acts as the architectural cornerstone. Within each stage, the components exhibit tight complementary roles: frequency-adaptive directional scanning captures geometric structures, 1×1 pointwise conv restores cross-channel correlations, low-frequency-guided gating mechanism suppresses high-frequency noise, and cross-orientation attention fusion resolves inter-frequency semantic conflicts. The full CFAS-Mamba integrates all components and achieves an OA of 96.35%, an AA of 90.09%, and a $\kappa \times 100$ of 95.83% on the IP dataset. Notably, the AA improves by 4.37 percentage points over the weakest ablated variant (*w/o Attn. Fusion*), underscoring the architecture's collective efficacy in HSIs classification.

F. Computational Complexity Analysis

To comprehensively evaluate the efficiency of the proposed method, Table VII reports the number of trainable parameters and floating-point operations (FLOPs) for all competing models. The measurement is standardized using an input patch size of 7×7 with 200 spectral bands at a batch size of 1.

TABLE VII
COMPUTATIONAL COMPLEXITY COMPARISON OF DIFFERENT MODELS

Category	Model	Params (K)	FLOPs (M)
CNN-based	M3D-DCNN	269.02	584.86
	DFFN	520.22	25.51
Transformer-based	SF	342.65	35.41
	SSFTT	931.85	19.06
Mamba-based	MambaHSI	69.02	1.85
	S^2Mamba	704.22	34.31
	Proposed	411.02	19.49

The analysis reveals distinct computational bottlenecks inherent to different architectural paradigms. Although the 3D-CNN model (M3D-DCNN) maintains a relatively lightweight parameter count (269.02K), it incurs a staggering computational cost of 584.86M FLOPs. This extreme overhead stems from the volumetric sliding operations of 3D convolutional kernels across both spatial and spectral dimensions, which fundamentally limits its deployment on resource-constrained platforms. Transformer-based architectures mitigate the spatial locality limitation but introduce substantial computational overhead. Specifically, SSFTT requires 931.85 K parameters, the highest among all competing methods, which is primarily attributed to the dense linear projections and multi-head self-attention mechanisms required for spatial-spectral tokenization. Although SF reduces the parameter count to 342.65 K, its FLOPs (35.41 M) remain prohibitively high due to the quadratic complexity $\mathcal{O}(N^2)$ of the attention matrix computation with respect to the sequence length. In contrast, models based on SSMs demonstrate a distinct advantage in computational efficiency owing to their linear complexity $\mathcal{O}(N)$. MambaHSI represents the extreme of this lightweight paradigm with merely 69.02K parameters and 1.85M FLOPs. However, as previously demonstrated in Tables II to V, this extreme compression causes severe spatial and spectral infor-

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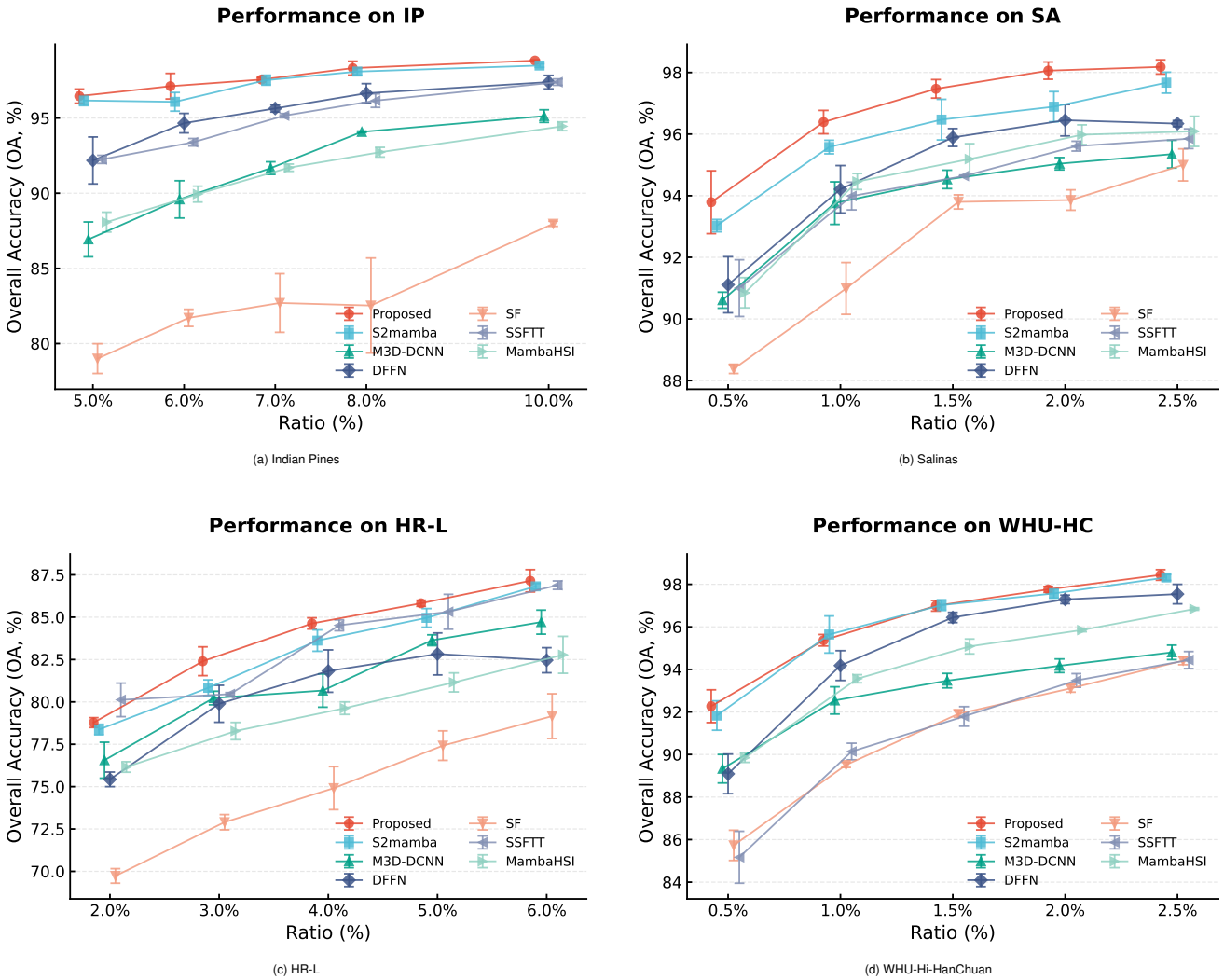


Fig. 7. Comparison of Overall Accuracy (OA) under different settings across four datasets. (a) IP. (b) SA. (c) HR-L. (d) WHU-HC.

mation loss, resulting in significantly degraded performance, particularly on heterogeneous datasets like HR-L. To recover accuracy, S^{2}Mamba employs bidirectional scanning mechanisms, but this dramatically inflates its complexity to 704.22K parameters and 34.31M FLOPs, losing the inherent efficiency advantage of SSMs.

The proposed CFAS-Mamba achieves an optimal trade-off between representation capacity and computational efficiency. Configured with 411.02 K parameters and 19.49 M FLOPs, the model demonstrates high efficiency: its computational cost is nearly 30-fold lower than that of M3D-DCNN and approximately 40% less than those of S^{2}Mamba and SF. Meanwhile, its parameter volume is less than half that of SSFTT.

G. Robustness Evaluation

To assess the robustness of CFAS-Mamba under limited supervision, we evaluate all methods across four datasets under varying training sample ratios. Specifically, the training ratios are set to $\{5\%, 6\%, 7\%, 8\%, 10\%\}$ for IP, $\{2\%, 3\%, 4\%, 5\%, 6\%\}$ for HR-L, and $\{0.5\%, 1\%, 1.5\%, 2\%, 2.5\%\}$ for SA and WHU-HC. Fig.7 visualizes the OA.

Across all evaluated settings, CFAS-Mamba consistently establishes a systematic lead over both CNN- and Transformer-based competitors. This performance advantage is most pronounced under extreme sample scarcity; at the lowest training ratios on WHU-HC and SA, CFAS-Mamba maintains an OA above 92%, substantially outpacing SSFTT, SF, and M3D-DCNN. This superior generalization capability demonstrates that the physically grounded frequency decomposition in our framework effectively extracts highly discriminative features even from minimal supervisory signals.

On the HR-L dataset, which features fragmented land-cover patches, the performance stratification across methods becomes sharper. While SSFTT holds a marginal advantage at the minimum labeling ratio, CFAS-Mamba and S^{2}Mamba rapidly close the gap and converge to the top tier as supervision increases. In contrast, DFFN and SF exhibit a pronounced performance plateau, failing to benefit proportionally from additional labeled samples. This indicates that frequency-aware serialization is intrinsically more adept at disentangling

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overlapping spectral features in heterogeneous terrains.

Furthermore, as the training ratios increase, almost all methods exhibit monotonic OA improvements, but with markedly different convergence behaviors. CFAS-Mamba shows rapid saturation on datasets with high land-cover homogeneity (SA and WHU-HC) while sustaining meaningful gains on the spectrally intricate IP and HR-L scenes. Notably, the standard deviation of CFAS-Mamba exhibits a consistent downward trend as supervision grows, reflecting enhanced numerical stability. While S^2Mamba yields comparably low variance at higher ratios, Transformer-based approaches show larger and less predictable fluctuations under low-label regimes, underscoring the brittleness of pure self-attention under data scarcity.

In summary, CFAS-Mamba demonstrates SOTA robustness under limited supervision, characterized by a decisive advantage at extremely low sampling rates, rapid convergence, and superior numerical stability across diverse hyperspectral scenes.

IV. Conclusion

In this study, a CFAS-Mamba network is proposed for HSI classification, which addresses the incompatibility between Mamba's 1D sequential input and 2D spatial-spectral topology through frequency-domain decoupling. Specifically, the WGAS module decomposes spatial-spectral features into physically oriented frequency subbands via channel-wise SWT and performs frequency-adaptive directional scanning, enabling lossless capture of directional edge details while preserving spectral correlation across embedding channels. Additionally, the CFSA module leverages the noise-robust low-frequency spectral features as a gating anchor to guide high-frequency components and employs cross-orientation attention fusion to reconstruct intact spectral-spatial textures. Stacked in a progressive hierarchical pyramid, these modules systematically extract multi-scale representations. Comprehensive experiments on four benchmark datasets demonstrate that CFAS-Mamba achieves superior classification accuracy compared to SOTA methods.

Nevertheless, the channel-wise SWT decomposition and multi-directional scanning convolution introduce extra computational overhead. Although the overall computational complexity outperforms 3D CNNs and conventional Transformers, the model retains more parameters and FLOPs compared with lightweight MambaHSI.

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